

ColdTrack AI — Evaluation Artifact (Baseline)

AIC COMPFEST 18 Final · Smart Logistics · tag **baseline** · system under test: MVP from the qualifying round (coldtrack.onnx GRU + coldtrack_ttb.onnx XGBoost + backend rule engine), evaluated **as deployed** on the held-out test split (16,244 sliding windows, 106 trips; split by trip).

Executive summary

Baseline score: **8/23 acceptance checks pass**. The headline model metrics from the qualifying round reproduce exactly (macro-F1 0.581, forecast MAE@30 0.198 °C), but the deployed system fails where it matters operationally:

#	Finding (evidence)	Severity
F1	1 in 3 imminent breaches gets a green status. End-to-end API, windows whose true time-to-breach ≤ 30 min: AMAN 32%, WASPADA 43%, KRITIS 25% (n=312).	Critical
F2	The TTB is silenced by the classifier. Of windows with true TTB ≤ 30 min, 68% are classified healthy (A0) and the backend hides TTB for A0; the two ONNX models show a TTB in only 25%.	Critical
F3	Fragile to realistic sensor noise. Adding $N(0, 0.3 \text{ °C})$ to temp_c (training noise was 0.05 °C): macro-F1 0.58 $\rightarrow$ 0.51; imminent breaches labelled healthy 68% $\rightarrow$ 82%.	High
F4	Stuck sensor while cargo warms is read as healthy. Only 10% flagged masalah_sensor; 73% predicted A0.	High
F5	Forecast target is met by a trivial baseline. 'Temperature stays the same' scores 0.195 °C vs GRU 0.198 °C; only 84 of 16,244 test windows move $\geq 1 \text{ °C}$ in 30 min.	High (metric validity)
F6	Three fault classes are largely undetected: recall A7 44%, A8 20%, degradasi_bertahap 11% (95% CI 0%–22%).	High
F7	Test set cannot support per-class or per-cargo claims. 2–4 test trips for A1/A2/A4/A5; the vaccine, frozen-meat and fish profiles have zero imminent-breach test windows.	Medium (validity)

1. Test suite: scope, cases, metrics and why

Design principle: test the system **the user actually sees** (raw ONNX outputs → backend gating → rule engine → HTTP response), not just the network. Code: `eval/` — `run_eval.py` (measurements), `test_baseline.py` (23 acceptance checks), `make_artifact.py` (this PDF). Reproduce: `python -m eval.run_eval --tag baseline; pytest eval/test_baseline.py`.

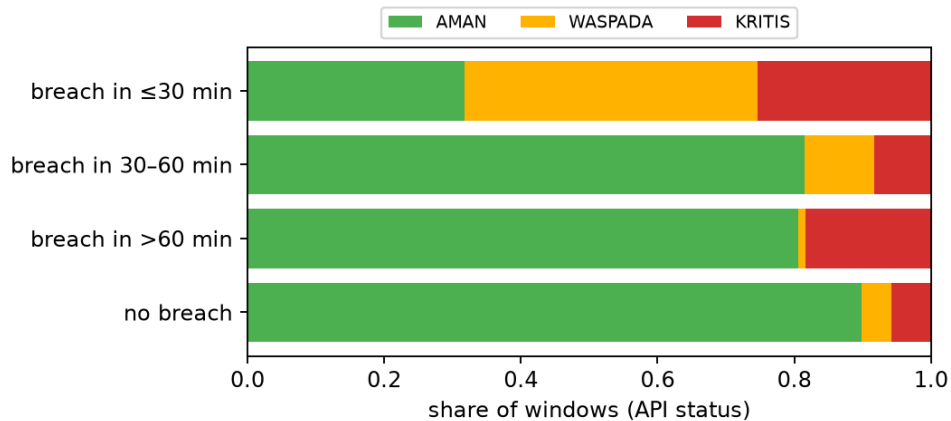
Group	Test cases	Metric(s)	Why this metric / what it does NOT capture
G1 Forecast	16,244 test windows; sliced by fault mode and cargo; compared to naive persistence and linear extrapolation	MAE t+15/30/60, p95 and max error, MAE on windows that really move	MAE is dominated by the ~99% of windows where nothing changes; hence the persistence comparison and the 'moving windows' slice. MAE says nothing about whether the alarm fires.
G2 Classification	7 classes; bootstrap over <i>trips</i> (200×) for CIs; recall vs minutes since fault onset	per-class P/R/F1, macro-F1, PR-AUC, confusion matrix	Macro-F1 hides which fault is missed → per-class recall checks. Windows of one trip are highly correlated, so window-level CIs would be far too narrow.
G3 Imminent breach	windows with true TTB ≤ 10 / ≤ 30 min, gated exactly like <code>backend/app/inference.py</code>	TTB shown-rate, MAE when shown, miss reasons (classifier said A0 / sensor / above cap)	The qualifying-round 'MAE 7.08 min for TTB ≤ 30 ' conditions on the true answer and ignores silent misses. Shown-rate + reason exposes them.
G4 Robustness	sensor noise $\sigma \in \{0.1, 0.3, 0.5\}$ °C (delta features recomputed); +7 h hour shift (UTC vs WIB); stuck sensor injected into imminent-breach windows	macro-F1, forecast MAE, imminent-breach A0 rate, class chosen under stuck sensor	Training data is a clean simulator ($\sigma=0.05$ °C, ACF too smooth per dataset card). Real loggers are noisier. Perturbations are synthetic, so they show <i>sensitivity</i> , not real-world error.
G5 End-to-end API	8,122 windows (every 2nd) POSTed to <code>/api/v1/analyze</code> via FastAPI TestClient, cargo profile mapped from the trip	status × true-TTB bucket, TTB shown, latency, parity with raw ONNX, HTTP errors	Ground truth = simulator's <code>time_to_breach</code> (threshold = cargo max limit; verified equal to <code>config.yaml</code>). Says nothing about real trucks.

Pre-existing checks (run for reference, not part of the graded suite): backend `pytest` **24/24 pass**, `ml/tests data-contract` **7/7 pass**. XGBoost/Linear/IsolationForest baselines (`ml/baselines.py`) re-run on the *test* split: forecast MAE@30 linear 0.192 / XGB 0.190; macro-F1 XGB 0.667; PR-AUC XGB 0.749, IsolationForest 0.352. The committed `baseline_metrics.json` is from the validation split of v3 and is stale.

Caveats we did not remove: 106 test trips (27 anomalous); all data synthetic; acceptance thresholds are our own operational choices, set with sight of the qualifying-round numbers and open to challenge.

2. Baseline failure findings (where the system fails, with evidence)

F1 · Imminent breaches shown as AMAN

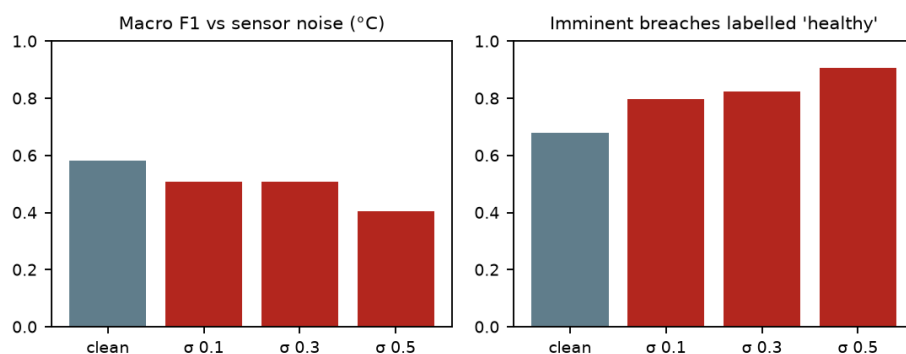


Root causes visible in code: (i) status is driven by the forecast head, which is $\sim$ persistence (F5); (ii) the TTB floor in rules.py only applies when a TTB exists, and it is None whenever the classifier says A0; (iii) the backend heuristic TTB fallback only runs after WASPADA/KRITIS. When shown, API TTB is accurate (MAE 1.6 min) — the failure is *coverage*, not precision. Also: only 19% of windows 30–60 min before breach are above AMAN, so the 60-min warning path in config.yaml is effectively unreachable (TTB display cap = 30).

F2 · TTB depends on the weakest component

True TTB ≤ 30 min (n=625)	Share
TTB shown by the two ONNX models + gate	25%
Hidden because classifier said A0 (healthy)	68%
Hidden because classifier said masalah_sensor	5%
Hidden because TTB model output ≥ 30 min cap	3%
MAE when shown / counting misses as '30 min'	3.4 min / 18.3 min

F3/F4 · Robustness



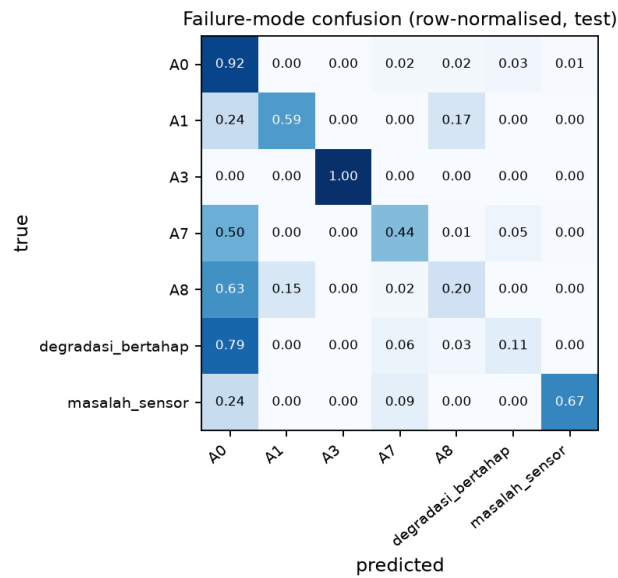
Timezone hypothesis tested and **refuted**: shifting hour_of_day by +7 h leaves macro-F1 at 0.588 (clean 0.581). Serving parity is good: API vs raw ONNX forecast mean $|\Delta|$ 0.005 $^{\circ}\text{C}$ (max 0.53 $^{\circ}\text{C}$), class disagreement 0.6% — no significant train/serve skew.

F5 • The forecast target is not discriminative

t+30 min MAE (°C)	All windows	Moving windows (n=84)
GRU (deployed)	0.198	1.24
Naive persistence ('same as now')	0.195	1.56
Linear extrapolation of last 5 min	0.614	2.65

The '<0.8 °C — target met' claim is true but uninformative. On the few windows where temperature changes, GRU beats persistence by ~20%, which is the honest headline.

F6 • Fault classes



Class	test windows	test trips	precision	recall	F1
A0	12546	95	0.87	0.92	0.90
A1	321	4	0.64	0.59	0.61
A3	379	7	1.00	1.00	1.00
A7	642	5	0.41	0.44	0.42
A8	651	6	0.30	0.20	0.24
degradasi_bertahap	833	2 (A2) + 3 (A4)	0.16	0.11	0.13
masalah_sensor	872	3 (A5) + 8 (A6)	0.89	0.67	0.76

Detection vs minutes since fault onset (any non-A0): 0-10 min: 56% (n=198); 10-30 min: 65% (n=300); 30-60 min: 76% (n=440); 60-120 min: 74% (n=925); 120-inf min: 38% (n=1835). Recall is *lowest deep into a fault* (120+ min) — so 'the fault has not shown yet' does not explain the misses. We have not yet established why (see §4).

F7 • Test-set validity

Imminent-breach ($TTB \leq 30$) test windows exist only for produce (319) and dairy (306); none for vaccine, frozen meat or fish — the flagship vaccine use case has no imminent-breach evidence. Bootstrap 95% CI for macro-F1 is 0.49–0.64.

3. Before / after iteration

Status: **baseline only**. This section is filled after each iteration (Evaluation Track); the 'after' column is intentionally empty until measured by the same suite.

Metric (test split)	Baseline	After iteration	Change / reason
Acceptance checks passed	8/23	—	—
Imminent breach shown as AMAN (API)	32%	—	—
Imminent breach: TTB shown (API)	62%	—	—
Imminent breach labelled A0 (models)	68%	—	—
Macro-F1	0.581	—	—
Recall degradasi / A8 / A7	0.11 / 0.20 / 0.44	—	—
Macro-F1 at $\sigma=0.3$ °C noise	0.509	—	—
Stuck sensor → flagged masalah_sensor	10%	—	—
Forecast MAE@30 on moving windows	1.24 °C	—	—

4. Findings not yet fixed — causes and plan

Causes below are **hypotheses** unless marked measured; each will be confirmed or rejected with the suite before any change.

Finding	Suspected cause (status)	Planned fix and how we will judge it
F1/F2 green status & silent TTB	TTB and status floors are gated on the classifier / forecast; both are weak on slow faults (measured: 68% of imminent windows → A0).	Add a forecast-independent guard: a physics/rate-based trigger from current temp vs cargo limit and recent slope, applying regardless of class. Judge: F1 AMAN-rate $\leq 5\%$ without raising healthy WASPADA+KRITIS above 10%.
F3 noise fragility	Trained only on $\sigma=0.05$ °C simulator noise (dataset card, measured). Aggregate features (std, trend) are noise-sensitive (hypothesis).	Noise augmentation at train time or smoothing before feature aggregation; retrain, re-run G4. Judge: F1 drop ≤ 0.05 at $\sigma=0.3$.
F4 stuck sensor	Frozen temp looks like a stable healthy trace (hypothesis). masalah_sensor recall only 67% even in-distribution (measured).	Explicit flat-line detector in preprocessing (zero variance over last N minutes while ambient/reefer state changes) as a rule, independent of the network.
F5 forecast metric	Metric choice, not a model defect (measured).	Report the moving-window MAE and skill vs persistence as the primary forecast metrics.
F6 A7/A8/degradasi	Label taken at window end; slow faults look healthy early (partly measured: recall is low even 120+ min in, so not only onset latency). Class overlap $A0 \leftrightarrow A8/A7$ /degradasi in confusion matrix (measured).	Inspect misclassified windows per class; consider longer context or class-specific features. May remain unresolved — we will report the residual.
F7 test validity	700 simulated trips; 106 in test (measured).	Regenerate a larger, cargo-stratified test set (data/only); report CIs. Cannot substitute for field data.

Appendix — acceptance checks (baseline)

Result	Check
PASS	test_forecast_mae_t30_under_0_8C
PASS	test_recall_per_fault_class_at_least_0_5[A1]
PASS	test_recall_per_fault_class_at_least_0_5[A3]
PASS	test_recall_per_fault_class_at_least_0_5[masalah_sensor]
PASS	test_timezone_shift_does_not_change_macro_f1_much
PASS	test_healthy_windows_not_over_alerted
PASS	test_api_no_http_errors_and_latency_p95_under_300ms
PASS	test_api_matches_raw_onnx_forecast
FAIL	test_forecast_worst_case_error_bounded
FAIL	test_macro_f1_at_least_0_80
FAIL	test_recall_per_fault_class_at_least_0_5[A7]
FAIL	test_recall_per_fault_class_at_least_0_5[A8]
FAIL	test_recall_per_fault_class_at_least_0_5[degradasi_bertahap]
FAIL	test_anomaly_recall_at_least_0_8
FAIL	test_imminent_breach_ttb_shown_directly_by_models_at_least_80pct
FAIL	test_imminent_breach_not_gated_as_healthy
FAIL	test_test_split_covers_every_cargo_profile_for_imminent_breach
FAIL	test_sensor_noise_0_3C_macro_f1_drop_at_most_0_05
FAIL	test_sensor_noise_0_3C_does_not_hide_imminent_breach
FAIL	test_stuck_sensor_during_breach_is_flagged_not_healthy
FAIL	test_imminent_breach_never_green_more_than_5pct
FAIL	test_imminent_breach_ttb_shown_by_api_at_least_80pct
FAIL	test_every_fault_mode_has_at_least_5_test_trips

Commit of the code that produced these numbers: a1b8a86 (+ uncommitted eval/ at generation time). Dataset: data/processed/v4_seed1000_700trips.parquet, split=test. Seeds fixed at 42.